

Answer Key — Core Mathematics Grade 10

Sub-Strand 2.4: Trigonometry 1 • Lesson 5: Trigonometric Ratios of Special Angles (45° , 30° , 60° , 0° , and 90°)

Teacher reference only. Do not project or hand out.

Quick marking grid

Q1	Q2	Q3	Q4	Q5	Q6	Q7
D	A	C	C	A	C	C

USE AFTER: OBSERVE PHASE — Construction A: isosceles right-angled triangle with legs of 6 cm, measuring BC

Q1. In Construction A, a group draws an isosceles right-angled triangle with both legs (AB and AC) 6 cm long. What is the exact length of the hypotenuse BC?

- A. $6\sqrt{3}$ cm (about 10.39 cm)
- B. 12 cm
- C. $\sqrt{6}$ cm (about 2.45 cm)
- D. $6\sqrt{2}$ cm (about 8.49 cm)**

Correct answer: D. By Pythagoras, $BC^2 = 6^2 + 6^2 = 72$, so $BC = \sqrt{72} = 6\sqrt{2} \approx 8.49$ cm. Adding the legs (12 cm) is the common error.

USE AFTER: OBSERVE PHASE — Construction B: equilateral triangle of side 8 cm bisected into a 30-60-90 triangle, measuring PM

Q2. In Construction B, a group draws an equilateral triangle PQR with side 8 cm and bisects it by drawing PM perpendicular to QR, where M is the midpoint of QR. What is the exact length of PM?

- A. $4\sqrt{3}$ cm (about 6.93 cm)**
- B. 12 cm
- C. $4\sqrt{2}$ cm (about 5.66 cm)
- D. 4 cm

Correct answer: A. $QM = 4$ cm and $QP = 8$ cm, so $PM^2 = 8^2 - 4^2 = 48$ and $PM = \sqrt{48} = 4\sqrt{3} \approx 6.93$ cm. This is longer than QM but shorter than QP.

USE AFTER: EXPLAIN PHASE — Part A: Exact derivation for the 45° triangle (legs = 1 unit, hypotenuse = $\sqrt{2}$)

Q3. An isosceles right-angled triangle has both legs 1 unit long and a hypotenuse of $\sqrt{2}$ units. What is the exact value of $\sin 45^\circ$?

- A. $\sqrt{2}$
- B. 1
- C. $1/\sqrt{2}$**
- D. $1/2$

Correct answer: C. $\sin 45^\circ = \text{opposite/hypotenuse} = 1/\sqrt{2}$. Choosing $\sqrt{2}$ comes from inverting the ratio, and 1 is the value of $\tan 45^\circ$, not $\sin 45^\circ$.

USE AFTER: EXPLAIN PHASE — Part A: Exact derivation for the 30-60-90 triangle (short leg = 1, long leg = $\sqrt{3}$, hypotenuse = 2)

Q4. A 30-60-90 triangle has sides 1 (opposite 30°), $\sqrt{3}$ (opposite 60°) and 2 (the hypotenuse). What is the exact value of $\tan 30^\circ$?

- A. $\sqrt{3}/2$
- B. $1/2$
- C. $1/\sqrt{3}$**
- D. $\sqrt{3}$

Correct answer: C. $\tan 30^\circ = \text{opposite/adjacent} = 1/\sqrt{3}$. Using $\sqrt{3}$ mixes up the sides, which gives $\tan 60^\circ$. $\sqrt{3}/2$ is $\sin 60^\circ$, and $1/2$ is $\sin 30^\circ$.

USE AFTER: EXPLAIN PHASE — Part B: Mast application (angle of elevation 60° , distance 40 m)

Q5. A surveyor in Nakuru stands 40 m from the base of a Kenya Power mast and measures the angle of elevation to the top as exactly 60° . Using $\tan 60^\circ = \sqrt{3}$, what is the height of the mast?

- A. $40\sqrt{3}$ m (about 69.3 m)
- B. 80 m
- C. 40 m
- D. $40/\sqrt{3}$ m (about 23.1 m)

Correct answer: A. height = $40 \times \tan 60^\circ = 40 \times \sqrt{3} = 40\sqrt{3} \approx 69.3$ m. Dividing by $\sqrt{3}$ comes from wrongly using $\tan 60^\circ = 1/\sqrt{3}$.

USE AFTER: EXPLAIN PHASE — Part A: looking back at the trig-table columns to complete the values for 0° and 90°

Q6. Why is $\tan 90^\circ$ described as undefined?

- A. $\tan 90^\circ = 1$, because the opposite side and the hypotenuse become equal
- B. $\tan 90^\circ = 0$, because the adjacent side becomes 0
- C. The adjacent side approaches 0, and division by zero is not possible
- D. The opposite side becomes 0, so there is nothing to divide

Correct answer: C. $\tan \theta = \text{opposite/adjacent}$. As the angle approaches 90° , the adjacent side approaches 0, and dividing by 0 has no answer, so $\tan 90^\circ$ is undefined.

USE AFTER: DQB PHASE — DQB phase: blue sticky note new question 'What if the angle is 50° — can we get an exact answer, or must we use tables?'

Q7. After today's lesson, a group asks: 'Can we find the exact height of the Nakuru mast if the angle of elevation is 50° ?' Which answer is correct?

- A. No, a mast can only be measured when the angle is 30° , 45° or 60°
- B. Yes, use the 60° triangle, because 50° is close to 60° , and the answer will be exact
- C. The two special triangles give exact values only for 0° , 30° , 45° , 60° and 90° , so for 50° we use tables or a calculator
- D. Yes, use the 45° triangle, because 50° is close to 45° , and the answer will be exact

Correct answer: C. Exact surd values come only from the special triangles. For a non-special angle such as 50° , we use tables or a calculator, which gives an approximate value.